

October 09, 2019

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50236921

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 26, 2019. The results relate only to the samples included in this report. Results reported herein conform to the most current, applicable TNI/NELAC standards and the laboratory's Quality Assurance Manual, where applicable, unless otherwise noted in the body of the report.

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids

Pace Project No.: 50236921

Indiana Certification IDs

7726 Moller Road, Indianapolis, IN 46268

Illinois Certification #: 200074

Indiana Certification #: C-49-06

Kansas/NELAP Certification #: E-10177

Kentucky UST Certification #: 80226

Kentucky WW Certification #: 98019

Michigan Department of Environmental Quality, Laboratory
#9050

Ohio VAP Certification #: CL0065

Oklahoma Certification #: 2018-101

Texas Certification #: T104704355

West Virginia Certification #: 330

Wisconsin Certification #: 999788130

USDA Soil Permit #: P330-16-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 50236921

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50236921001	Digester #4	Solid	09/25/19 13:10	09/26/19 09:50

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50236921

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50236921001	Digester #4	EPA 6010	KJE	6	PASI-I
		EPA 6020	DMT	10	PASI-I
		EPA 7471	ILP	1	PASI-I
		SM 2540G	SKK	1	PASI-I
		SM 2540G	SKK	1	PASI-I
		SM 2540G	SKK	1	PASI-I
		EPA 9045	WDB	1	PASI-I
		EPA 351.2	DAC1	1	PASI-I
		EPA 353.2	DAC1	2	PASI-I
		SM 4500-NH3 G	MMS	1	PASI-I
		SM 4500-CI-E	NMW	1	PASI-I

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50236921

Sample: Digester #4 **Lab ID: 50236921001** Collected: 09/25/19 13:10 Received: 09/26/19 09:50 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
6010 MET ICP Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Aluminum	9570	mg/kg	2020	1	09/29/19 09:23	10/01/19 11:07	7429-90-5	
Calcium	36700	mg/kg	2020	1	09/29/19 09:23	10/01/19 11:07	7440-70-2	
Magnesium	6140	mg/kg	2020	1	09/29/19 09:23	10/01/19 11:07	7439-95-4	
Phosphorus	52000	mg/kg	2020	1	09/29/19 09:23	10/01/19 11:07	7723-14-0	N2
Potassium	3250	mg/kg	2020	1	09/29/19 09:23	10/01/19 11:07	7440-09-7	
Sodium	5610	mg/kg	2020	1	09/29/19 09:23	10/01/19 11:07	7440-23-5	
6020 MET ICPMS Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Arsenic	19.6	mg/kg	0.87	1	09/30/19 10:30	10/01/19 09:11	7440-38-2	
Cadmium	1.7	mg/kg	0.44	1	09/30/19 10:30	10/01/19 09:11	7440-43-9	
Chromium	106	mg/kg	1.7	1	09/30/19 10:30	10/01/19 09:11	7440-47-3	
Copper	903	mg/kg	4.4	10	09/30/19 10:30	10/02/19 00:11	7440-50-8	
Lead	30.1	mg/kg	4.4	5	09/30/19 10:30	10/01/19 07:22	7439-92-1	
Molybdenum	26.5	mg/kg	0.87	1	09/30/19 10:30	10/01/19 09:11	7439-98-7	N2
Nickel	24.8	mg/kg	0.44	1	09/30/19 10:30	10/01/19 09:11	7440-02-0	
Selenium	4.9	mg/kg	0.87	1	09/30/19 10:30	10/01/19 09:11	7782-49-2	
Silver	2.5	mg/kg	0.44	1	09/30/19 10:30	10/01/19 09:11	7440-22-4	
Zinc	1410	mg/kg	43.5	10	09/30/19 10:30	10/02/19 00:11	7440-66-6	
7471 Mercury Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Mercury	<1.7	mg/kg	1.7	1	10/02/19 20:22	10/03/19 10:47	7439-97-6	
Percent Moisture Analytical Method: SM 2540G								
Percent Moisture	97.7	%	0.10	1		09/28/19 13:39		
2540G Total Percent Solids Analytical Method: SM 2540G								
Total Solids	2.3	%	0.10	1		09/28/19 13:39		N2
2540G Total Volatile Solids Analytical Method: SM 2540G								
Total Volatile Solids	52.5	%	0.10	1		09/28/19 13:39		N2
9045 pH Soil Analytical Method: EPA 9045								
pH at 25 Degrees C	8.0	Std. Units	0.10	1		09/30/19 10:06		H3
351.2 Total Kjeldahl Nitrogen Analytical Method: EPA 351.2 Preparation Method: EPA 351.2								
Nitrogen, Kjeldahl, Total	71500	mg/kg	2090	1	10/02/19 15:30	10/04/19 11:01	7727-37-9	N2
353.2 Nitrogen, NO2/NO3 Analytical Method: EPA 353.2 Preparation Method: EPA 353.2								
Nitrogen, NO2 plus NO3	<214	mg/kg	214	1	10/08/19 15:19	10/09/19 11:56		N2
Nitrogen, Nitrate	<214	mg/kg	214	1	10/08/19 15:19	10/09/19 11:56	14797-55-8	N2
4500 Ammonia Soil, Distilled Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B								
Nitrogen, Ammonia	42100	mg/kg	2190	1	10/03/19 12:50	10/03/19 15:51	7664-41-7	N2

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50236921

Sample: Digester #4 **Lab ID: 50236921001** Collected: 09/25/19 13:10 Received: 09/26/19 09:50 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E								
Chloride	7620	mg/kg	4290	1	10/01/19 12:03	10/04/19 11:25	16887-00-6	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 524604

Analysis Method: EPA 7471

QC Batch Method: EPA 7471

Analysis Description: 7471 Mercury

Associated Lab Samples: 50236921001

METHOD BLANK: 2421083

Matrix: Solid

Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.19	0.19	10/03/19 10:20	

LABORATORY CONTROL SAMPLE: 2421084

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.48	0.48	100	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2421085 2421086

Parameter	Units	50237083017 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.58	0.62	0.58	0.61	98	98	75-125	6	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50236921

QC Batch:	524062	Analysis Method:	EPA 6010
QC Batch Method:	EPA 3050	Analysis Description:	6010 MET
Associated Lab Samples:	50236921001		

METHOD BLANK: 2418582 Matrix: Solid
Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	09/30/19 16:42	
Calcium	mg/kg	<50.0	50.0	09/30/19 16:42	
Magnesium	mg/kg	<50.0	50.0	09/30/19 16:42	
Phosphorus	mg/kg	ND	50.0	09/30/19 16:42	N2
Potassium	mg/kg	<50.0	50.0	09/30/19 16:42	
Sodium	mg/kg	<50.0	50.0	09/30/19 16:42	

LABORATORY CONTROL SAMPLE: 2418583

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	508	102	80-120	
Calcium	mg/kg	500	499	100	80-120	
Magnesium	mg/kg	500	499	100	80-120	
Phosphorus	mg/kg	50	52.8	106	80-120	N2
Potassium	mg/kg	500	493	99	80-120	
Sodium	mg/kg	500	489	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2418584 2418585

Parameter	Units	50236926001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	300	475	423	799	739	105	104	75-125	8	20	
Calcium	mg/kg	7520	475	423	8270	8060	158	127	75-125	3	20	P6
Magnesium	mg/kg	491	475	423	933	893	93	95	75-125	4	20	
Phosphorus	mg/kg	14800	47.5	42.3	15600	14800	1620	-88	75-125	5	20	E, N2, P6
Potassium	mg/kg	964	475	423	1550	1520	123	131	75-125	2	20	M0
Sodium	mg/kg	446	475	423	1010	983	119	127	75-125	3	20	M0

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50236921

QC Batch: 524191	Analysis Method: EPA 6020
QC Batch Method: EPA 3050B	Analysis Description: 6020 MET
Associated Lab Samples: 50236921001	

METHOD BLANK: 2419470 Matrix: Solid
Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	10/01/19 06:41	
Cadmium	mg/kg	<0.050	0.050	10/01/19 06:41	
Chromium	mg/kg	<0.20	0.20	10/01/19 06:41	
Copper	mg/kg	<0.050	0.050	10/01/19 06:41	
Lead	mg/kg	<0.10	0.10	10/01/19 06:41	
Molybdenum	mg/kg	<0.10	0.10	10/01/19 06:41	N2
Nickel	mg/kg	<0.050	0.050	10/01/19 06:41	
Selenium	mg/kg	<0.10	0.10	10/01/19 06:41	
Silver	mg/kg	<0.050	0.050	10/01/19 06:41	
Zinc	mg/kg	<0.50	0.50	10/01/19 06:41	

LABORATORY CONTROL SAMPLE: 2419471

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.8	95	80-120	
Cadmium	mg/kg	4	3.9	98	80-120	
Chromium	mg/kg	4	4.1	104	80-120	
Copper	mg/kg	4	4.0	101	80-120	
Lead	mg/kg	4	4.1	104	80-120	
Molybdenum	mg/kg	4	4.2	106	80-120	N2
Nickel	mg/kg	4	4.0	100	80-120	
Selenium	mg/kg	4	3.9	97	80-120	
Silver	mg/kg	4	4.1	103	80-120	
Zinc	mg/kg	4	3.8	95	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2419472 2419473

Parameter	Units	50236438009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	13.8	5.1	5.1	32.9	30.6	373	330	75-125	7	20	M3
Cadmium	mg/kg	0.30	5.1	5.1	4.8	4.8	88	89	75-125	1	20	
Chromium	mg/kg	18.5	5.1	5.1	22.8	24.3	84	115	75-125	7	20	
Copper	mg/kg	21.6	5.1	5.1	24.1	32.2	50	211	75-125	29	20	1d,P6
Lead	mg/kg	18.6	5.1	5.1	34.7	30.6	315	237	75-125	12	20	P6
Molybdenum	mg/kg	<3.1	5.1	5.1	6.7	6.7	104	105	75-125	0	20	N2
Nickel	mg/kg	50.5	5.1	5.1	39.0	45.8	-225	-92	75-125	16	20	P6
Selenium	mg/kg	<3.1	5.1	5.1	7.9	8.5	101	113	75-125	7	20	
Silver	mg/kg	<0.063	5.1	5.1	4.7	4.7	91	91	75-125	0	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2419472 2419473												
Parameter	Units	50236438009 Result	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max	
			Spike Conc.	Spike Conc.							RPD	Qual
Zinc	mg/kg	102	5.1	5.1	102	104	-9	41	75-125	2	20	P6

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 524000

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: Dry Weight/Percent Moisture

Associated Lab Samples: 50236921001

SAMPLE DUPLICATE: 2418106

Parameter	Units	50236921001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	97.7	97.7	0	5	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 523998

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Solids

Associated Lab Samples: 50236921001

SAMPLE DUPLICATE: 2418100

Parameter	Units	50236921001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	2.3	2.3	1	10	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 523999

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Associated Lab Samples: 50236921001

SAMPLE DUPLICATE: 2418101

Parameter	Units	50236921001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	52.5	51.8	1	10	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 524208

Analysis Method: EPA 9045

QC Batch Method: EPA 9045

Analysis Description: 9045 pH

Associated Lab Samples: 50236921001

SAMPLE DUPLICATE: 2419517

Parameter	Units	50237050001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	2.7	2.8	1	2	H3

SAMPLE DUPLICATE: 2419518

Parameter	Units	50237083017 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.6	7.6	0	2	H3

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 524818

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Associated Lab Samples: 50236921001

METHOD BLANK: 2422107

Matrix: Solid

Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.0	49.0	10/04/19 10:59	N2

LABORATORY CONTROL SAMPLE: 2422108

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	490	454	93	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2422109 2422110

Parameter	Units	50236921001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	71500	21400	20500	90000	93500	86	107	90-110	4	20	M0, N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50236921

QC Batch:	525797	Analysis Method:	EPA 353.2
QC Batch Method:	EPA 353.2	Analysis Description:	353.2 Nitrate + Nitrite
Associated Lab Samples:	50236921001		

METHOD BLANK: 2426414 Matrix: Solid
Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	10/09/19 11:53	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	10/09/19 11:53	N2

LABORATORY CONTROL SAMPLE: 2426415

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.6	107	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.9	20.8	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2426416 2426417

Parameter	Units	50237814001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	8.7	11.8	11.8	22.1	22.3	114	115	90-110	1	20	N2
Nitrogen, NO2 plus NO3	mg/kg	9.8	23.6	23.6	35.2	35.5	108	109	90-110	1	20	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 524943

Analysis Method: SM 4500-NH3 G

QC Batch Method: SM 4500-NH3 B

Analysis Description: 4500 Ammonia, Distilled

Associated Lab Samples: 50236921001

METHOD BLANK: 2422631

Matrix: Solid

Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	10/03/19 15:44	N2

LABORATORY CONTROL SAMPLE: 2422632

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	20	20.8	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2422633 2422634

Parameter	Units	50236479001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	30400	3250	3180	35100	34800	145	138	90-110	1	20	N2,P6

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50236921

QC Batch: 524482

Analysis Method: SM 4500-Cl-E

QC Batch Method: SM 4500-Cl-E

Analysis Description: 4500 Chloride

Associated Lab Samples: 50236921001

METHOD BLANK: 2420461

Matrix: Solid

Associated Lab Samples: 50236921001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.6	99.6	10/04/19 11:23	N2

LABORATORY CONTROL SAMPLE: 2420462

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	200	100	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2420463 2420464

Parameter	Units	50236921001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	7620	8590	8590	16800	16700	106	106	90-110	0	20	N2

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50236921

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

LABORATORIES

PASI-I Pace Analytical Services - Indianapolis

ANALYTE QUALIFIERS

1d	RPD value is outside control limits due to sample non-homogeneity.
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H3	Sample was received or analysis requested beyond the recognized method holding time.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50236921

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50236921001	Digester #4	EPA 3050	524062	EPA 6010	524334
50236921001	Digester #4	EPA 3050B	524191	EPA 6020	524374
50236921001	Digester #4	EPA 7471	524604	EPA 7471	524939
50236921001	Digester #4	SM 2540G	524000		
50236921001	Digester #4	SM 2540G	523998		
50236921001	Digester #4	SM 2540G	523999		
50236921001	Digester #4	EPA 9045	524208		
50236921001	Digester #4	EPA 351.2	524818	EPA 351.2	525208
50236921001	Digester #4	EPA 353.2	525797	EPA 353.2	525987
50236921001	Digester #4	SM 4500-NH3 B	524943	SM 4500-NH3 G	525061
50236921001	Digester #4	SM 4500-CI-E	524482	SM 4500-CI-E	524503

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WM#: 50236921

CHAIN-OF-CUSTODY / Analytical Request Doc
The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be c

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be c

Section A		Section B		Section C	
Required Client Information:		Required Project Information:		Invoice Information:	
Company:	City of Cadillac	Report To:	Cindy Tomaszewski	Attention:	
Address:	1121 Plett Rd	Copy To:		Company Name:	
	Cadillac, MI 49601			Address:	
Email:	ctomaszewski@cadillac-mi.net	Purchase Order #:		Pace Quote:	
Phone:	231-775-2388	Project Name:	Fall Blossolds	Pace Project Manager:	
Fax:		Project #:		Pace Profile #:	7875
Requested Due Date:					

[illegible]

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>9.20.19 11:00</u>		Evaluated by: <u>WDC</u>		WO#: 50236921 PM: MSB Due Date: 10/10/19 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>		Profile ID:			
Project Manager: <u>MSB</u>					
Rush TAT Requested: YES ☒ NO ☒ Due Date:					
Lab Notified of Rush or Short Holds: YES ☒ NO ☒		Non Conformance Form Required: YES ☐ NO ☒			
Samples Received Via: FedEx ☒ UPS ☒ Client ☐ Pace Courier ☐ Other: _____					Comments:
Custody Seals Present and Intact:		YES ☒	NO ☐	NA ☐	
Received Sample Information Form(s): Drinking Waters Only		YES ☒	NO ☐	NA ☐	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES ☒	NO ☐	N/A ☒	
Short Holds Present (< 72 Hours):		YES ☒	NO ☐	☒	
Samples Received in Hold:		YES ☒	NO ☐	☒	
Custody Signatures Present:		YES ☒	NO ☐	☒	
Collector Signature Present:		YES ☒	NO ☐	☒	
Packing Material Used:		YES ☒	NO ☐	☒	
Samples Collected Today and On Ice:		YES ☒	NO ☐	N/A ☒	
IR Gun #: 280 ☒ 281 ☒		Digital Thermometer #: 282 ☒ 283 ☒			
Ice Type: WET Bagged / WET Loose ☒ BLUE ☐ NONE ☐		1. Cooler Temp Upon Receipt: <u>3.3/3.9</u> °C			
Ice Location: TOP ☒ BOTTOM ☐ MIDDLE ☐ DISPERSED ☐		Temp should be 0-6°C (Initial/Corrected)			
Temp Blank Received:		YES ☒	NO ☐	☒	
Containers Intact:		YES ☒	NO ☐	☒	
Correct Containers:		YES ☒	NO ☐	☒	
Sufficient Volume:		YES ☒	NO ☐	☒	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl		YES ☒	NO ☐	N/A ☒	
Residual Chlorine Absent: (SVOC/Pest 625, PCB 608, Total/Amenable/Available Cyanide)		YES ☒	NO ☐	N/A ☒	
VOA Headspace Acceptable (<6mm):		YES ☒	NO ☐	N/A ☒	
Trip Blank Received: HCl ☐ MeOH ☐ TSP ☐ OTHER ☐		YES ☒	NO ☐	☒	
Comments:		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			